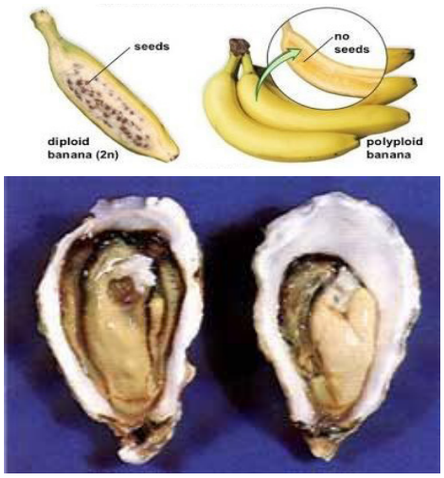


Figure 18.10 Euploidy

Aneuploidy: It is the **loss** or **gain** of **one** or **more chromosomes** in a set. It is of three types. **Monosomy** (2n-1), **Trisomy** (2n+1)and **Nullisomy** (2n-2). In man, Down's syndrome is one of the commonly known aneuploid condition.

Down's syndrome

This condition was first identified by a doctor named **Langdon Down** in1866.

It is a genetic condition in which there is an extra copy of **chromosome 21 (Trisomy 21)**. It is associated with mental retardation, delayed development, behavioural problems, weak muscle tone, vision and hearing disability are some of the conditions seen in these children.

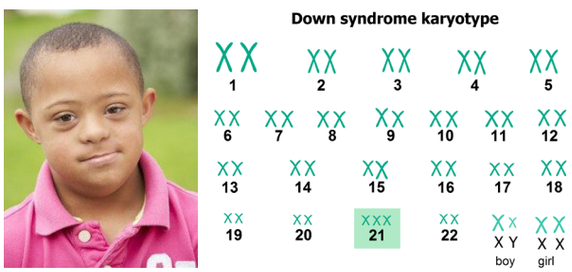


Figure 18.11 Aneuploidy

2. Gene or point mutation

Gene mutation is the **changes** occurring in **nucleotide sequence of a gene**. It involves substitution, deletion, insertion or inversion of a single or more than one nitrogenous base. Gene alteration results in abnormal protein formation in an organism.

Sickle cell anaemia is caused by the mutation of a single gene. Alteration in the gene brings a change in the structure of the protein part of haemoglobin molecule. Due to the change in the protein molecule, the red blood cell (RBC) that carries the haemoglobin is sickle shaped.

Points to Remember

- Variations are quite apparent among closely related groups of organisms.
- Mendel had choosen seven pairs of distinguishing traits: flower colour, position, seed colour, shape, pod colour, pod shape, stem length.
- Every pea plant has two 'factors' which are responsible for producing a particular character or trait is called allele.
- The process of acquiring characters or traits from parents is called 'Heredity'.
- Each human cell contains 23 pairs of chromosomes. Out of these 22 pairs are called autosomes and one pair is called allosomes.
- A chromosome consists of the regions: primary constriction, centromere, secondary constriction, telomere and satellite.
- Based on the position of the centromere, the chromosomes are classified as telocentric, acrocentric, sub metacentric and metacentric chromosomes.
- Each nucleotide of DNA consists of a deoxyribose sugar, a nitrogenous base and a phosphate group. Pairing is always between a purine and a pyrimidine.
- The sperm, produced by the father, determines the sex of the child. The mother is not responsible in determining the sex of the child.
- Mutation is an inheritable change in the genetic material of an organism.

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